

ARUL RHIK MAZUMDER

U.S. Citizen

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Research interests. I study what quantum algorithms really cost and when they are worth it: making algorithms such as Hamiltonian simulation cheaper on fault-tolerant hardware, proving how many T gates and ancilla qubits any circuit must use, and understanding when quantum approaches can genuinely outperform classical algorithms.

Education

Imperial College London

MSc, Physics with Quantum Dynamics

Sep. 2026 – Sep. 2027 (expected)

London, UK

Carnegie Mellon University

B.S. in Computer Science; ML and Algorithms & Complexity concentrations

Aug. 2023 – May 2026

Pittsburgh, PA

- Graduated with University Honors. Relevant coursework: Quantum Computing, Complexity Theory, Learning Theory, Randomized Algorithms, Convex Optimization, Parallel Computer Architecture.

Research Experience

Institute for Quantum Computing (IQC), University of Waterloo

Undergraduate Researcher (Mitacs Globalink Scholar), hosted by Michele Mosca and Vlad Gheorghiu

May 2026 – Aug. 2026

Waterloo, ON, Canada

- Developed tighter T-count bounds for structured unitaries: polynomial-time lower bounds for diagonal Clifford+T gates, computed from the gate's phase polynomial and exact for layers of disjoint Toffoli-type gates ($6m + 1$ T gates) in the {CNOT, T} model with clean ancillas. Single-author preprint; presented at QRE 2026 and UnitaryCon 2026.
- Paired the floor with an $O(n^3)$ upper bound that brackets every gate; the bounds carry over to Toffoli networks and finite-field multiplication oracles, recovering the optimal 17 T gates of the GF(4) multiplier by exact decoding.
- Released [code and data](#) that rerun every computational claim in the paper with one command.
- Completed USEQIP: quantum information coursework plus 30+ hours of hands-on experiments on IQC's quantum hardware.

Caltech Institute for Quantum Information and Matter (IQIM)

SURF Undergraduate Researcher, Preskill group, with Samson Wang and James Watson (Google Quantum AI)

May 2025 – Aug. 2025

Pasadena, CA

- Worked on early fault-tolerant algorithms for matrix functions, using commutator scaling to lower the cost of product-formula routines; written up as [arXiv:2608.13862](https://arxiv.org/abs/2608.13862) and presented as a poster at TQC 2026.
- Joined a biweekly reading group on quantum signal processing (QSP) and quantum singular value transformation (QSVT).

BlueQubit

Part-time Researcher and Hackathon Lead, AI and Core teams

Oct. 2025 – Present

Los Angeles, CA (remote)

- Benchmarked GPU-accelerated MPS and Pauli-path simulators against AWS Braket, Quantum Rings and open-source backends; the runtime scaling study was released as a [preprint](#) and informed product decisions and simulator cost estimators.
- Lead student and community engagement: turn BlueQubit research into accessible, technically accurate blog posts and tutorials, and mentor students and judge the BlueQubit track at student quantum hackathons.

Quantum Technologies Group, Carnegie Mellon University

Researcher with Sridhar Tayur

May 2023 – Jun. 2025

Pittsburgh, PA

- Wrote a tutorial paper with five [notebooks](#) on solving QUBO problems with QAOA and quantum annealing, and tested semidefinite programs for entanglement detection.

Algorithms Group, George Mason University

Researcher with Fei Li

May 2022 – Jun. 2025

Remote

- Benchmarked quantum string matching, a hybrid D-Wave algorithm for the traveling salesman problem, and quantum annealing for shipment rerouting against classical solvers on real transport networks.

Publications

Quantum algorithms and complexity

1. **A. R. Mazumder.** “Exact T-Counts of Toffoli Layers from an Isotropy Bound.” Preprint, 2026; [code and data](#). Proves that a layer of m disjoint Toffoli gates needs exactly $6m + 1$ T gates, using a polynomial-time lower bound.
2. **A. R. Mazumder, J. Watson, S. Wang.** “Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling.” [arXiv:2608.13862](https://arxiv.org/abs/2608.13862); TQC 2026, poster. Estimates matrix functions of a Hamiltonian from plain Trotter circuits with one ancilla, at a depth set by nested commutators, through randomized extrapolation.

3. J. Lee, **A. R. Mazumder**. “Robust Adaptive T-Count Lower Bounds for Boolean Oracles.” Manuscript in preparation, 2026. Extends T-count lower bounds to adaptive circuits that may err; inner product and majority still need $\Theta(n)$ T gates. It also bounds the clean workspace that T-efficient synthesis of Boolean oracles requires.

Applied quantum computing and benchmarking

1. **A. R. Mazumder**. “An End-to-End Hybrid Quantum–Classical Sampling Workflow for Discrete Markov Random Fields.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026.
2. **A. R. Mazumder**, S. R. Mazumder. “Depth-Efficient Quantum Topological Data Analysis for Regime-Specific Detection of Financial Stress.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026.
3. **A. R. Mazumder** et al. “Benchmarking Zero-Setup Quantum Circuit Simulators.” arXiv:2607.09882, 2026. Runtime scaling of GPU-accelerated MPS and Pauli-path simulators against cloud and open-source backends.
4. **A. R. Mazumder**, S. Tayur. “Five Starter Problems: Solving Quadratic Unconstrained Binary Optimization Models on Quantum Computers.” *Tutorials in Operations Research*, INFORMS, 2025, pp. 145–183.
5. Earlier work benchmarking quantum optimization against classical solvers, with F. Li, A. Sen, U. Sen and others, 2022–2025: *IPDPS 2023*, *LNNS 2024*, *JSSR 2024*, *AISS 2022*, and arXiv:2501.05624 on quantum annealing for shipment rerouting.

Talks and Posters

- **UnitaryCon 2026**, Toronto — poster, “Certified T-count floors for Toffoli networks and reversible circuits.”
- **Quantum Resource Estimation (QRE) 2026**, Toronto — poster, same title.
- **IEEE QCE 2026**, Toronto — talks on two first-author papers, on hybrid quantum–classical sampling for discrete Markov random fields and on depth-efficient quantum topological data analysis of financial stress.
- **TQC 2026** — poster, “Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling” ([PDF](#)).
- **Quantum Leap Career Nexus**, University of Maryland, Nov. 2025, and **Caltech IQIM SURF Seminar**, Aug. 2025 — talks, “Early Fault-Tolerant Quantum Algorithms for Matrix Functions via Trotter Extrapolation” ([slides](#)).
- **INFORMS Annual Meeting 2025** — [presentation](#) on solving QUBO models on quantum computers.

Awards and Honors

- **IEEE QCE Student Travel Grant** (2026), IEEE Quantum Technical Community.
- **Mitacs Globalink Research Award and IQC Undergraduate Research Award** (2026), funding the IQC research term.
- **Duke STAQ NSF-funded Invited Scholar** (2025): week-long PhD-level seminar on quantum simulation, error correction and architecture with Peter Love, Kenneth Brown and Yu Tong, including a Duke Quantum Center lab tour.
- **MIT iQuHACK** (2026): 2nd, NVIDIA track, of 90+ teams; variational LABS with Pauli correlation encodings on GPUs.
- **YQuantum** (2025, Yale): Yale Quantum Institute Grand Prize; 1st place, Alice & Bob track; 2nd overall; Wigner-function quantum state reconstruction via convex optimization and convolutional-network denoising.
- **Citadel/Correlation One Datathon** (2024): 2nd of 150 invited from 3,000+ applicants; pairs trading, Sharpe 1.58.
- **HackCMU** (2023): Graduate Student Association Infrastructure Prize for Eco-Bin, an AI waste sorter piloted with CMU Dining.
- **Regeneron Science Talent Search Scholar** (2023), top 300 of 2,000+ applicants nationally.
- **Olympiads**: USA(J)MO qualifier, USAPhO Bronze, USACO Gold, USNCO National Finalist.

Leadership

CMU Quantum

Sep. 2024 – May 2026

Founder and President

Pittsburgh, PA

- Founded CMU’s first student quantum organization and grew it to 200+ members and 10+ student leaders; secured sponsorship from General Motors and Boeing; ran workshops with IBM, IonQ and Quantinuum.
- Developed teaching materials ([slides](#) and [handouts](#)), led hackathon tracks, and organized sponsored research trips.
- Ran on-campus seminars and tutorials in quantum information science, and led student teams to national hackathons.

Technical Skills

Languages and proof tools: Python, Julia, C++, Lean 4

Quantum and optimization: Qiskit, PennyLane, CUDA-Q, D-Wave Ocean; CVXPY, PICOS, Gurobi; CUDA, MPI